

Lecture 5: Motion Along a Curve & Arc Length

Background & Motivation

Last time we defined vector-valued functions and their derivatives. Today we put them to work: the derivative $\mathbf{r}'(t)$ gives velocity, its magnitude gives speed, and integrating the speed gives arc length. We also derive the projectile motion equations you may have seen in physics—but now with the full vector calculus treatment.

1. Kinematics of Space Curves

Definition 1: Position, Velocity, Acceleration

For a particle moving along $\mathbf{r}(t)$:

Position:	$\mathbf{r}(t)$	where the particle is
Velocity:	$\mathbf{v}(t) = \mathbf{r}'(t)$	direction and rate of movement
Speed:	$\ \mathbf{v}(t)\ = \ \mathbf{r}'(t)\ $	how fast (scalar!)
Acceleration:	$\mathbf{a}(t) = \mathbf{v}'(t) = \mathbf{r}''(t)$	rate of change of velocity

Recovering Position from Velocity/Acceleration

$$\mathbf{v}(t) = \int \mathbf{a}(t) dt + \mathbf{C}_1 \quad \mathbf{r}(t) = \int \mathbf{v}(t) dt + \mathbf{C}_2$$

– Use initial conditions $\mathbf{v}(0)$ and $\mathbf{r}(0)$ to find the constants

- **Displacement** (net change): $\int_a^b \mathbf{v}(t) dt$ – a vector
- **Total distance traveled:** $\int_a^b \|\mathbf{v}(t)\| dt$ – a scalar (always $\geq$ displacement magnitude)

Example 1: Displacement vs. Distance

A particle has velocity $\mathbf{v}(t) = \langle 3t, 4, t^2 \rangle$ on $[0, 2]$.

- Find the displacement.
- Set up the integral for total distance traveled.

(a) Displacement:

1. Integrate each component:

$$\int_0^2 \mathbf{v}(t) dt = \left\langle \left[\frac{3t^2}{2} \right]_0^2, [4t]_0^2, \left[\frac{t^3}{3} \right]_0^2 \right\rangle = \left\langle 6, 8, \frac{8}{3} \right\rangle$$

(b) Total distance:

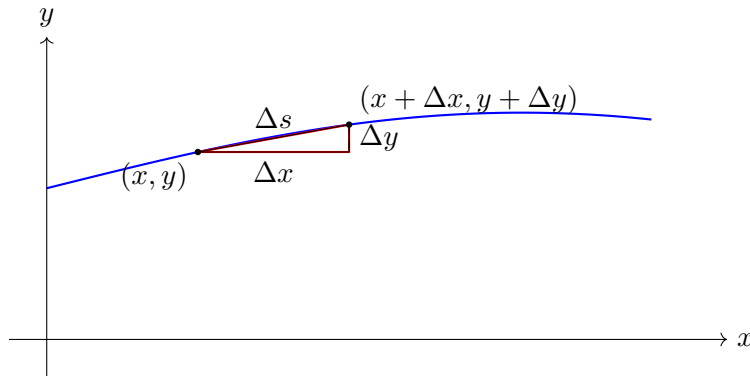
1. Speed: $\|\mathbf{v}(t)\| = \sqrt{9t^2 + 16 + t^4}$
2. Integral: $\int_0^2 \sqrt{9t^2 + 16 + t^4} dt$ – no nice closed form, use numerical methods.

2. Arc Length

Theorem 1: Arc Length of a Parametric Curve

Idea. Approximate the curve by a chain of tiny straight segments. Each segment between $\mathbf{r}(t)$ and $\mathbf{r}(t + dt)$ has length $\|\mathbf{r}'(t)\| dt$ (velocity times time). Adding these up in the limit gives the total arc length:

$$L = \int_a^b \|\mathbf{r}'(t)\| dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$



Approximating the arc length of a curve using a right triangle with Δs as the hypotenuse, Δx as the base, and Δy as the height. Summing Δs over infinitely many such triangles gives the integral formula for L .

Arc Length in Other Coordinates

Cartesian ($y = f(x)$):

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Polar ($r = f(\theta)$):

$$L = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

Example 2: Arc Length of a Helix

Find the arc length of $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 3t \rangle$ for $t \in [0, 2\pi]$.

1. Derivative: $\mathbf{r}'(t) = \langle -2 \sin t, 2 \cos t, 3 \rangle$

2. Speed: $\|\mathbf{r}'(t)\| = \sqrt{4 \sin^2 t + 4 \cos^2 t + 9} = \sqrt{4 + 9} = \sqrt{13}$

3. Arc length: $L = \int_0^{2\pi} \sqrt{13} dt = \boxed{2\pi\sqrt{13}}$

Note: the speed is constant – the helix is traversed at a uniform rate.

Example 3: Arc Length of a t^2/t^3 Curve

Find the arc length of $\mathbf{r}(t) = \langle t^2, t^3 \rangle$ for $t \in [0, 1]$.

1. Derivative: $\mathbf{r}'(t) = \langle 2t, 3t^2 \rangle$

2. Speed: $\|\mathbf{r}'(t)\| = \sqrt{4t^2 + 9t^4} = t\sqrt{4 + 9t^2}$

3. Arc length:

$$L = \int_0^1 t\sqrt{4 + 9t^2} dt$$

4. Substitution: $u = 4 + 9t^2$, $du = 18t dt$:

$$L = \frac{1}{18} \int_4^{13} \sqrt{u} du = \frac{1}{18} \cdot \frac{2}{3} \left[u^{3/2} \right]_4^{13} = \frac{1}{27} (13\sqrt{13} - 8) = \boxed{\frac{13\sqrt{13} - 8}{27}}$$

3. Projectile Motion

– Projectile launched from origin, initial speed v_0 , angle θ above horizontal, no air resistance

Projectile Motion Equations

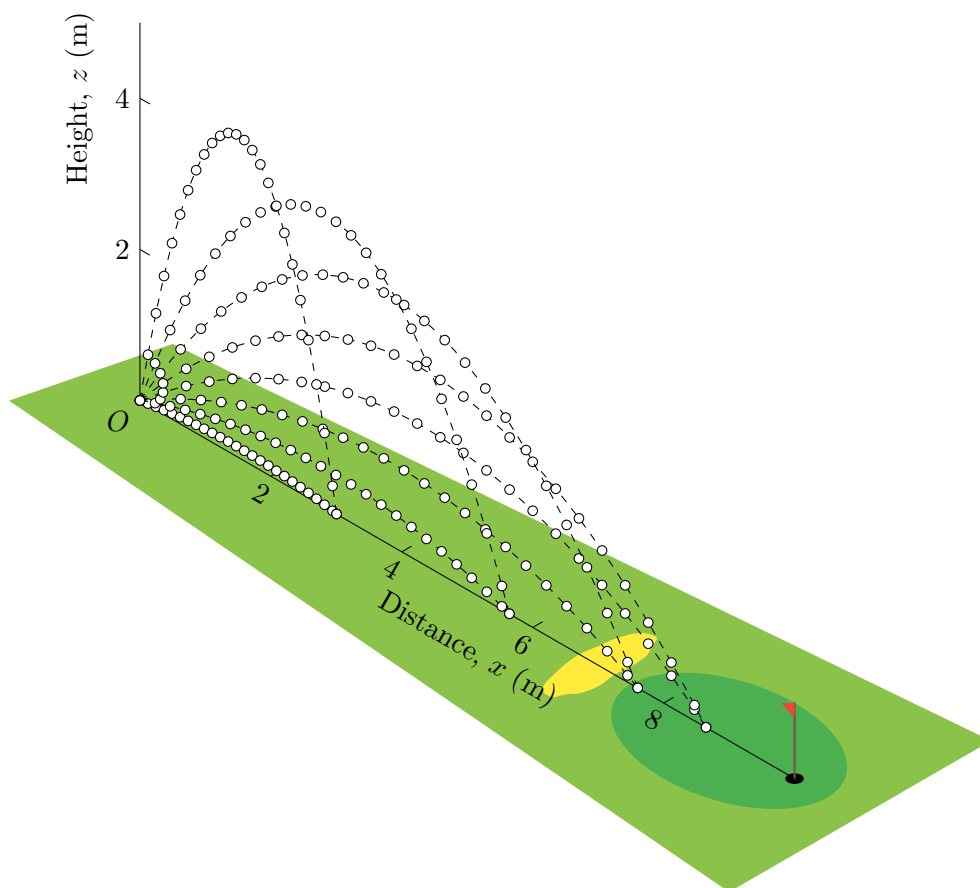
$$\mathbf{r}(t) = \langle v_0 \cos \theta \cdot t, v_0 \sin \theta \cdot t - \frac{1}{2}gt^2 \rangle$$

$$\text{Max height: } H = \frac{v_0^2 \sin^2 \theta}{2g}$$

$$\text{Time of flight: } T = \frac{2v_0 \sin \theta}{g}$$

$$\text{Range: } R = \frac{v_0^2 \sin(2\theta)}{g}$$

– Max range at $\theta = 45^\circ$ – Max height at $\theta = 90^\circ$



Projectile trajectories at several launch angles, shown on a golf course. In the absence of air resistance each ball follows a parabolic path; the 45° launch gives the longest range.

Example 4: Projectile Problem

A ball is launched at $v_0 = 20$ m/s at $\theta = 45^\circ$. Find max height, time of flight, and range. ($g = 9.8$ m/s²)

1. Setup: $\sin 45^\circ = \cos 45^\circ = \frac{\sqrt{2}}{2}$
2. Max height: $H = \frac{20^2 \cdot (1/2)}{2(9.8)} = \frac{200}{19.6} \approx \boxed{10.2 \text{ m}}$
3. Time of flight: $T = \frac{2(20)(\sqrt{2}/2)}{9.8} = \frac{20\sqrt{2}}{9.8} \approx \boxed{2.89 \text{ s}}$
4. Range: $R = \frac{400 \sin 90^\circ}{9.8} = \frac{400}{9.8} \approx \boxed{40.8 \text{ m}}$

Note: $\theta = 45^\circ$ maximizes range since $\sin(2 \cdot 45^\circ) = 1$.

Show all work for full credit.

Practice Problems

Problem 1. Find the velocity, speed, and acceleration of $\mathbf{r}(t) = \langle \sqrt{t}, t^3, e^t \rangle$ at $t = 1$.

Answer: $\mathbf{v}(1) = \langle 1/2, 3, e \rangle$, speed = $\sqrt{37/4 + e^2}$, $\mathbf{a}(1) = \langle -1/4, 6, e \rangle$

Problem 2. Find the arc length of $\mathbf{r}(t) = \langle t, t^2/\sqrt{2}, t^3/3 \rangle$ for $t \in [0, 1]$.

Hint: simplify $\|\mathbf{r}'(t)\|$ – it factors.

Answer: $L = 4/3$

Problem 3. Find the arc length of the cardioid $r = 1 + \cos \theta$ for $\theta \in [0, 2\pi]$.

Hint: use $1 + \cos \theta = 2 \cos^2(\theta/2)$.

Answer: $L = 8$

Problem 4. A golfer hits a ball at $v_0 = 45$ m/s. What angle gives maximum range, and what is that range?

Answer: $\theta = 45^\circ$, $R = \frac{2025}{9.8} \approx 206.6$ m

Problem 5. A particle has $\mathbf{a}(t) = \langle 0, 0, -g \rangle$, $\mathbf{v}(0) = \langle 3, 4, 10 \rangle$ m/s, $\mathbf{r}(0) = \langle 0, 0, 100 \rangle$ m. Find $\mathbf{r}(t)$ and the time it hits the ground ($z = 0$).

Answer: $\mathbf{r}(t) = \langle 3t, 4t, -\frac{g}{2}t^2 + 10t + 100 \rangle$

Time: $t = \frac{10 + \sqrt{2060}}{9.8} \approx 5.65$ s